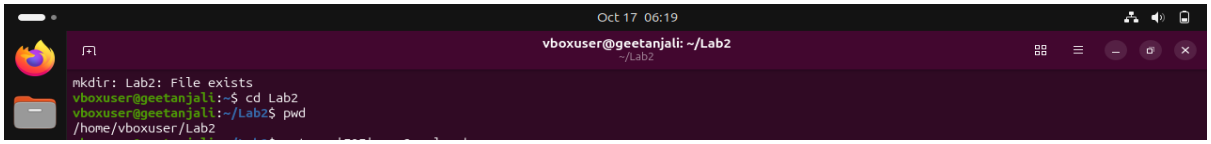
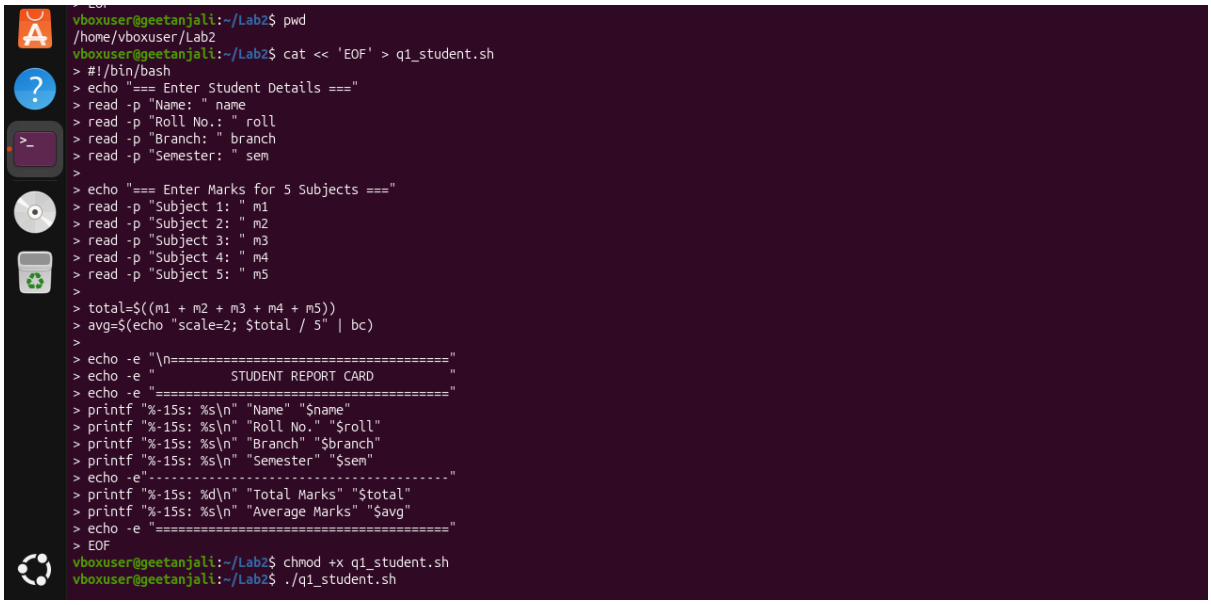
LAB ASESSMENT 2

QUETSION-1

Student Information System script accepting Name, Roll No., Branch, Semester and marks in five subjects using variables; display a formatted report.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or messages on the terminal.
- read -p – displays a prompt and stores the user's input in a variable.
- variables – store values. Use $name to access the value stored in name.
- $((…)) – performs integer arithmetic. Here, it calculates total and average.



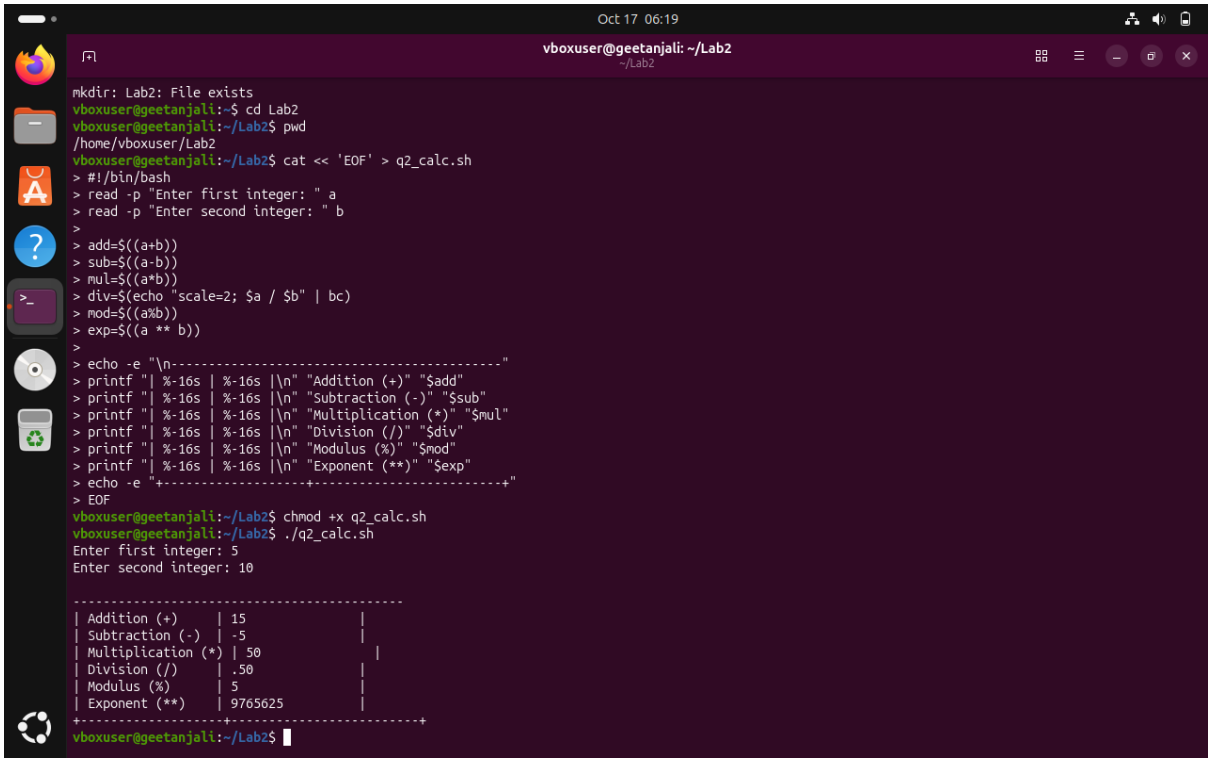



vboxuser@geetanjali:~/Lab2$ chmod +x q1_student.sh
vboxuser@geetanjali:~/Lab2$ ./q1_student.sh
=== Enter Student Details ===
Name: Geetanjali
Roll No.: 0003
Branch: CS
Semester: 1st
=== Enter Marks for 5 Subjects ===
Subject 1: 90
Subject 2: 88
Subject 3: 97
Subject 4: 83
Subject 5: 99
STUDENT REPORT CARD
Name : Geetanjali
Roll No. : 0003
Branch : CS
Semester : 1st
-e
Total Marks : 457
Average Marks : 91.40
vboxuser@geetanjali:~/Lab2$

QUESTION 2.

Script accepting two integers and performing Addition, Subtraction, Multiplication, Division, Modulus and Exponentiation; display results in tabular format.

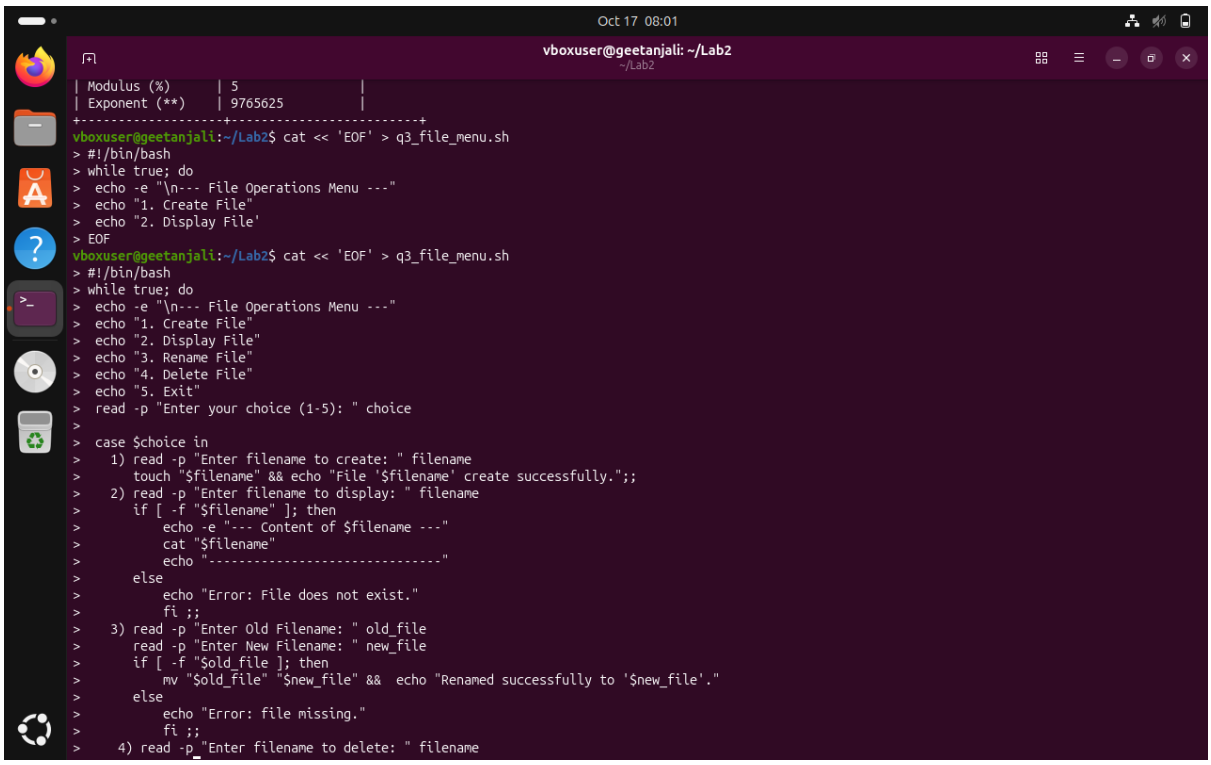
- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or messages on the terminal.
- read -p – reads the two numbers into variables a and b.
- if…the…else…fi – runs one block if a condition is true and another if it is false.
- [“$b” -ne 0] – tests whether b is not equal to zero. -ne means ‘not equal’ for integer comparisons.

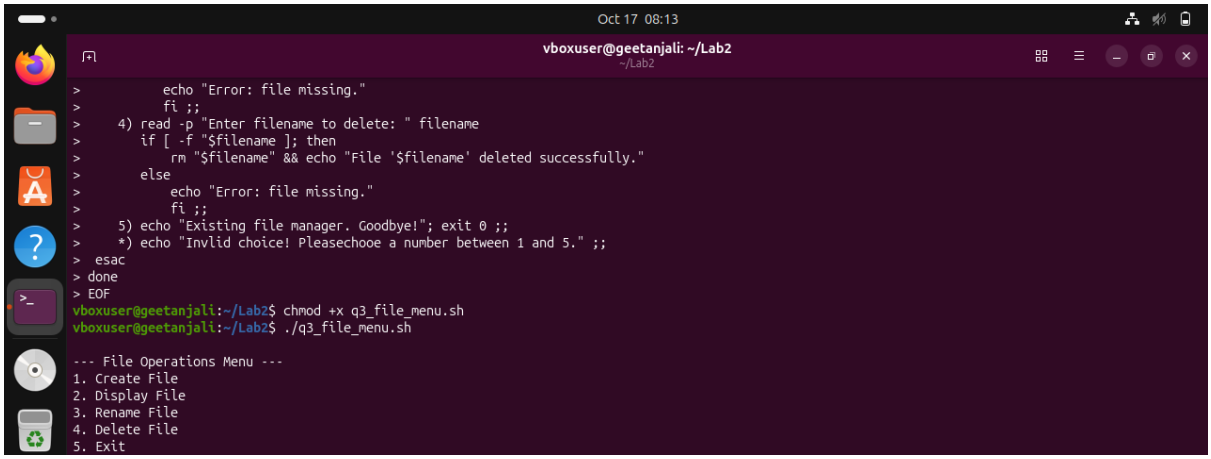


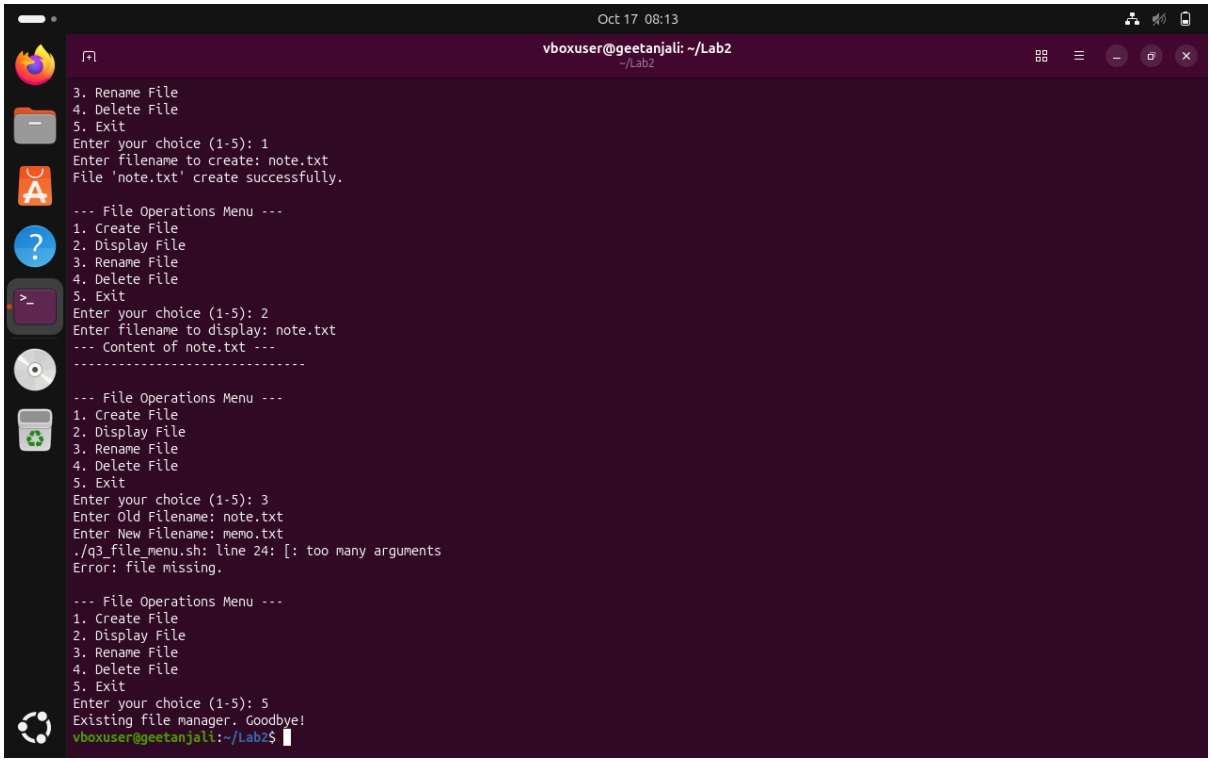
QUESTION 3.

Menu-driven script (case statement) for Create/Display/Rename/Delete File and Exit, with invalid choice handling.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or messages on the terminal.
- case… in … esac – chooses the matching action based on the user's menu choice.
- [-e "$f"] – true if a path exists. Used to avoid creating over an existing path.
- touch – "$f" – creating an empty file if it does not already exist, -- marks the end of options.
- cat --"$f" – display the contents of a file.
- mv -- "$old" "$new" renames or moves a file.
- rm -- "$f" – deletes a file. Use carefully because deletion may be irreversible.
- && - runs the command on the right only .




Oct 17 08:13
vboxuser@geetanjali: ~/Lab2
~/Lab2
>   echo "Error: file missing."
>   fi ;;
>   4) read -p "Enter filename to delete: " filename
>   if [-f "$filename]; then
>   rm "$filename" && echo "File '$filename' deleted successfully."
>   else
>   echo "Error: file missing."
>   fi ;;
>   5) echo "Existing file manager. Goodbye!"; exit 0 ;;
>   *) echo "Invlid choice! Pleasechooe a number between 1 and 5." ;;
> esac
> done
> EOF
vboxuser@geetanjali:~/Lab2$ chmod +x q3_file_menu.sh
vboxuser@geetanjali:~/Lab2$ ./q3_file_menu.sh
--- File Operations Menu ---
1. Create File
2. Display File
3. Rename File
4. Delete File
5. Exit

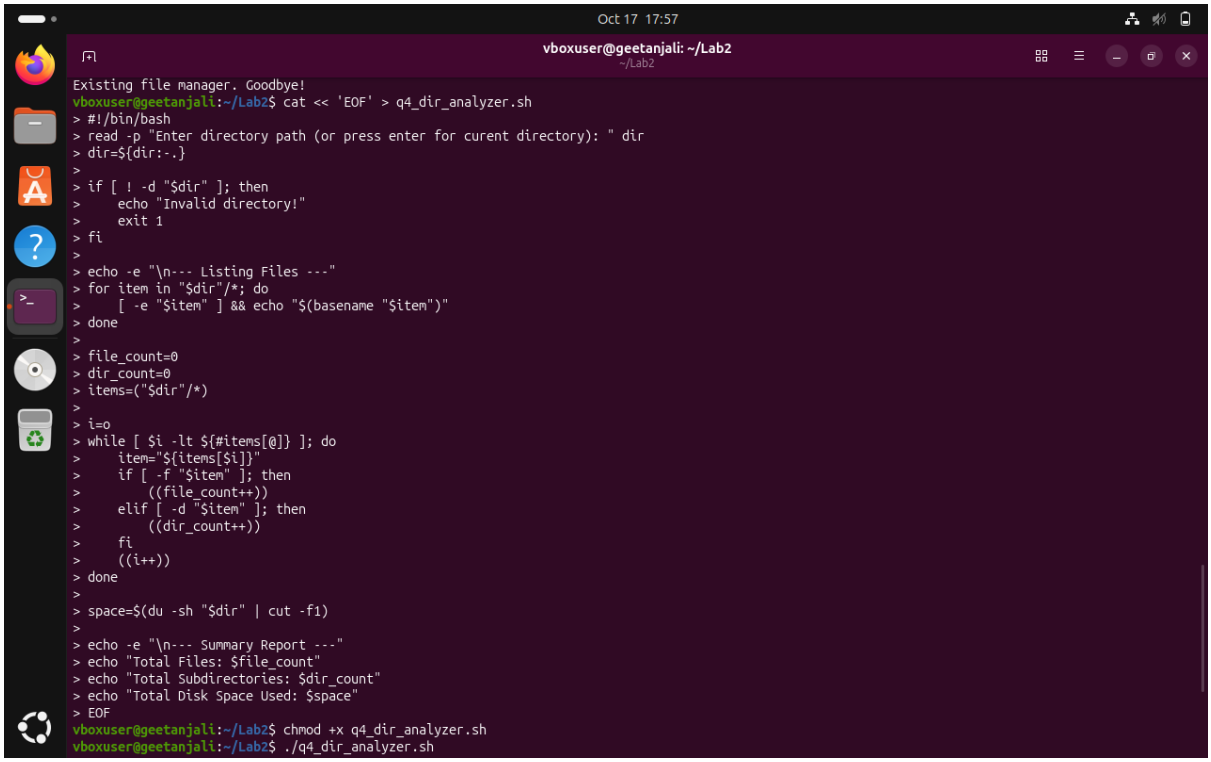

Oct 17 08:13
vboxuser@geetanjali: ~/Lab2
~/Lab2
3. Rename File
4. Delete File
5. Exit
Enter your choice (1-5): 1
Enter filename to create: note.txt
File 'note.txt' create successfully.
--- File Operations Menu ---
1. Create File
2. Display File
3. Rename File
4. Delete File
5. Exit
Enter your choice (1-5): 2
Enter filename to display: note.txt
--- Content of note.txt ---

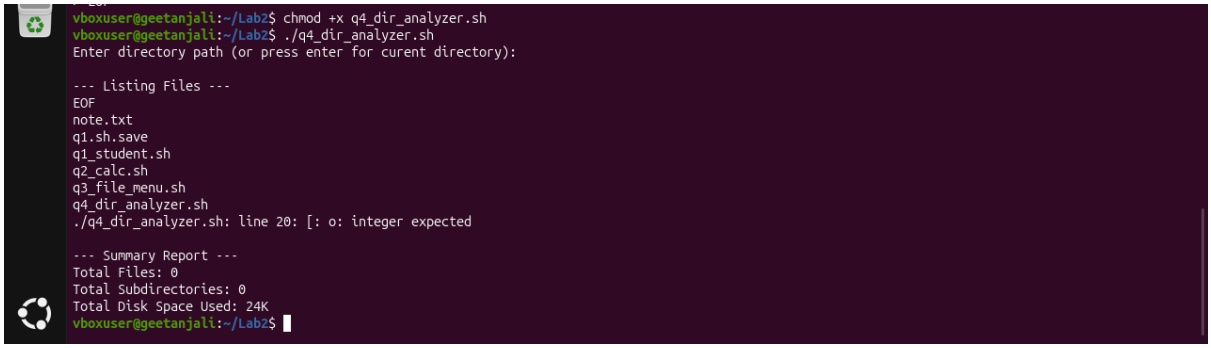
--- File Operations Menu ---
1. Create File
2. Display File
3. Rename File
4. Delete File
5. Exit
Enter your choice (1-5): 3
Enter Old Filename: note.txt
Enter New Filename: memo.txt
./q3_file_menu.sh: line 24: [: too many arguments
Error: file missing.
--- File Operations Menu ---
1. Create File
2. Display File
3. Rename File
4. Delete File
5. Exit
Enter your choice (1-5): 5
Existing file manager. Goodbye!
vboxuser@geetanjali:~/Lab2$

QUESTION 4.

Script using for and while loops to list files in a directory, count files/subdirectories, and calculate disk used.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or messages on the terminal.
- [-e “$item”] || continue – if the item does not exist, continue skips to the next loop item. || means OR / run the next commands if the first fails.
- [-f] and [-d] – test whether an entry is a regular file or a directory.
- [! -d “$dir”] – checks whether the supplied path is not a directory . ! means logical NOT; -d checks a directory.

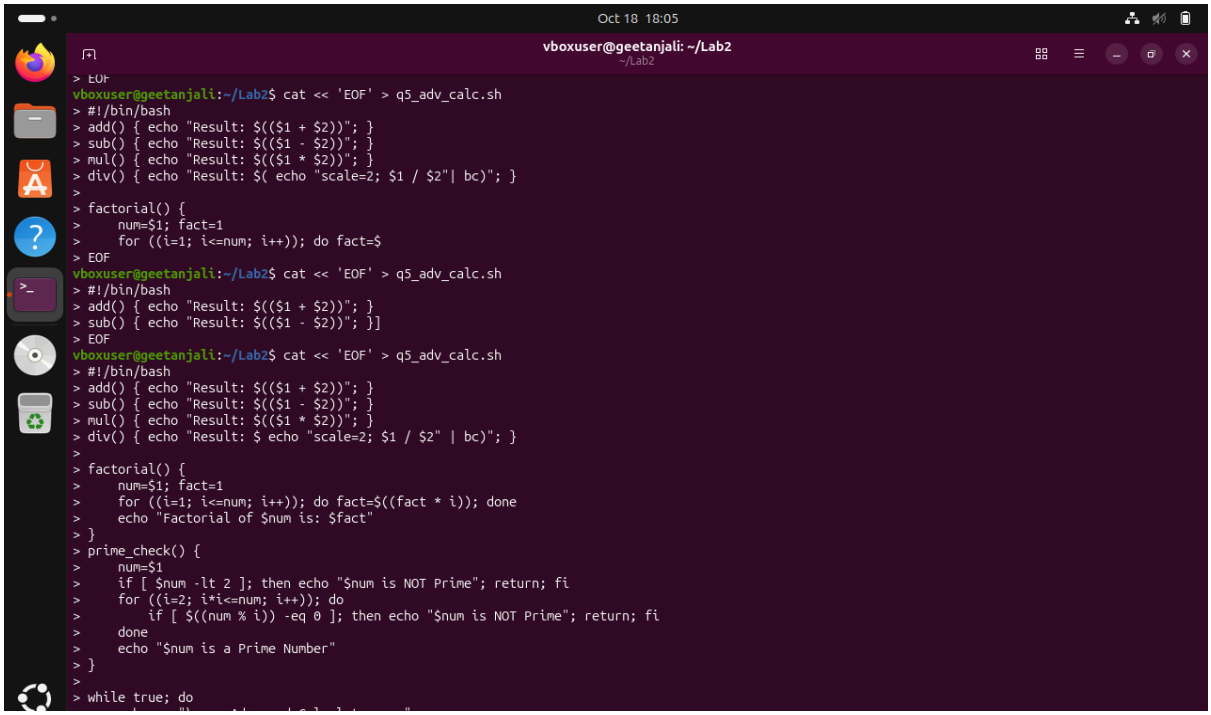


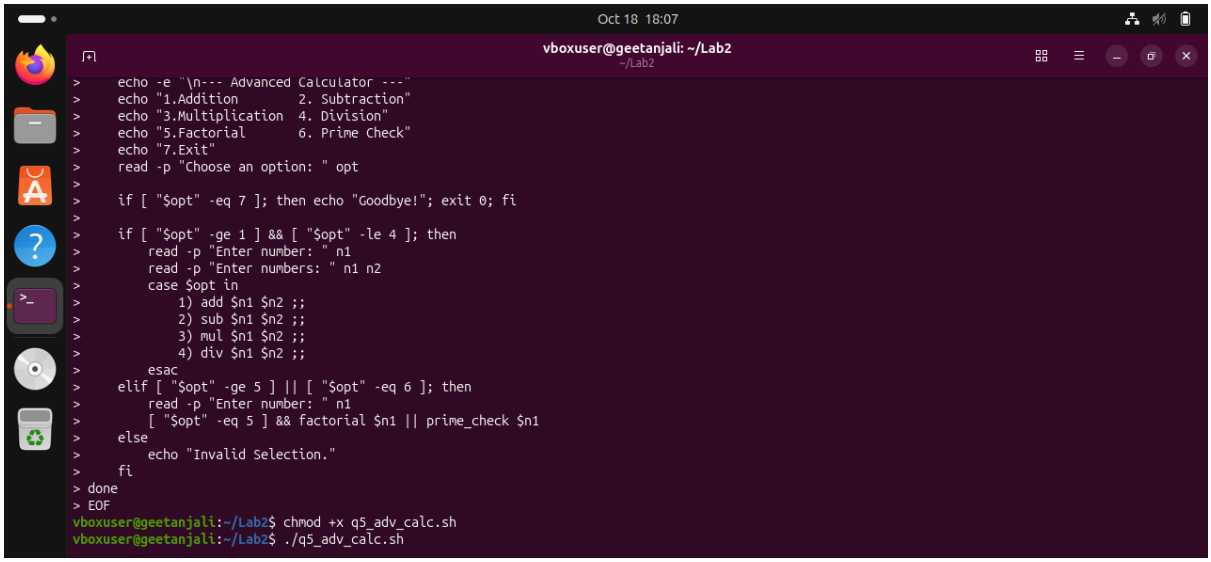

vboxuser@geetanjali:~/Lab2$ chmod +x q4_dir_analyzer.sh
vboxuser@geetanjali:~/Lab2$ ./q4_dir_analyzer.sh
Enter directory path (or press enter for curent directory):
--- Listing Files ---
EOF
note.txt
q1.sh.save
q1_student.sh
q2_calc.sh
q3_file_menu.sh
q4_dir_analyzer.sh
./q4_dir_analyzer.sh: line 20: [: o: integer expected
--- Summary Report ---
Total Files: 0
Total Subdirectories: 0
Total Disk Space Used: 24K
vboxuser@geetanjali:~/Lab2$

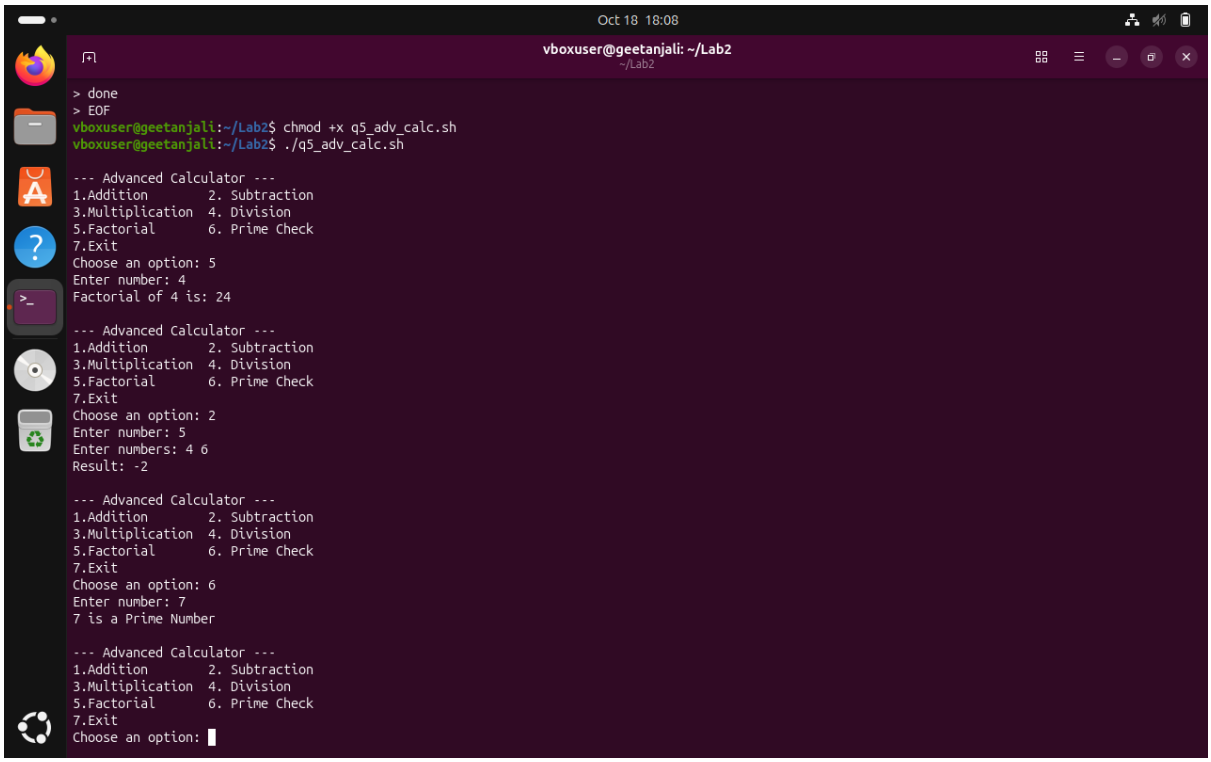
QUESTION 5.

Calculator application using shell functions (Addition, Subtraction, Multiplication, Division, Factorial, Prime Check) with a repeating menu.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or message on the terminal.
- read – displays the prompt and stores user's input in a variable.
- -eq, -lt, -le – integer comparisons: equal, less than, and less than or equal to.
- case – routes the chosen menu number to the relevant function.
- return – stops the current function. It does not exit the whole script.




Oct 18 18:07
vboxuser@geetanjali: ~/Lab2
~/Lab2
> echo -e "\n--- Advanced Calculator ---"
> echo "1.Addition 2. Subtraction"
> echo "3.Multiplication 4. Division"
> echo "5.Factorial 6. Prime Check"
> echo "7.Exit"
> read -p "Choose an option: " opt
>
> if ["$opt" -eq 7]; then echo "Goodbye!"; exit 0; fi
>
> if ["$opt" -ge 1] && ["$opt" -le 4]; then
> read -p "Enter number: " n1
> read -p "Enter numbers: " n1 n2
> case $opt in
> 1) add $n1 $n2 ;;
> 2) sub $n1 $n2 ;;
> 3) mul $n1 $n2 ;;
> 4) div $n1 $n2 ;;
> esac
> elif ["$opt" -ge 5] || ["$opt" -eq 6]; then
> read -p "Enter number: " n1
> ["$opt" -eq 5] && factorial $n1 || prime_check $n1
> else
> echo "Invalid Selection."
> fi
> done
> EOF
vboxuser@geetanjali:~/Lab2$ chmod +x q5_adv_calc.sh
vboxuser@geetanjali:~/Lab2$ ./q5_adv_calc.sh

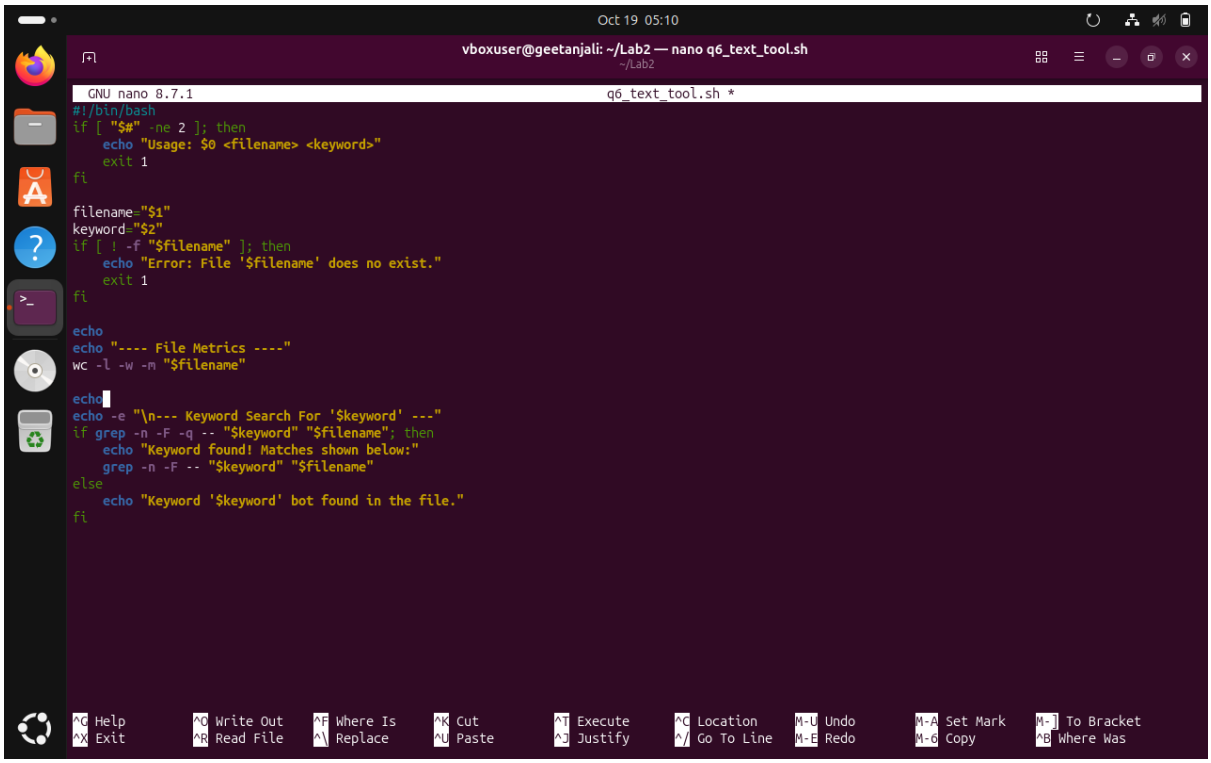

Oct 18 18:08
vboxuser@geetanjali: ~/Lab2
~/Lab2
> done
> EOF
vboxuser@geetanjali:~/Lab2$ chmod +x q5_adv_calc.sh
vboxuser@geetanjali:~/Lab2$ ./q5_adv_calc.sh
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 5
Enter number: 4
Factorial of 4 is: 24
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 2
Enter number: 5
Enter numbers: 4 6
Result: -2
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 6
Enter number: 7
7 is a Prime Number
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option:

QUESTION 6.

Script accepting filename & keyword as command-line arguments; verify file existence, display line/word/character counts and search for the keyword.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or message on the terminal.
- read – displays the prompt and stores user's input in a variable.
- $# - special variable is script name; $1 is the first argument (filename); $2 is the second argument (keyword).
- exit 1 – stops the script and returns a non-zero status, conventionally indicating an error.
- wc -l – counts newline characters/line.
- wc -m – counts characters according to the current locale; includes newline characters.





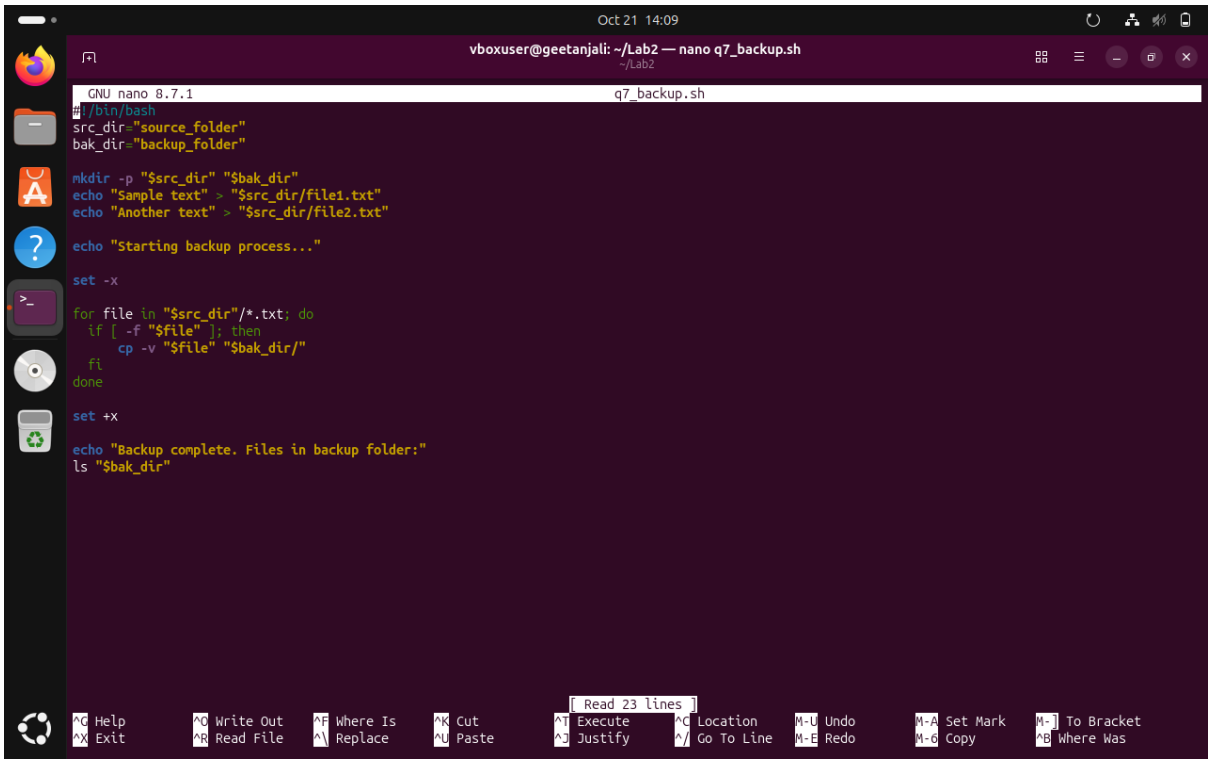

2:hello bash script coding
vboxuser@geetanjali:~/Lab2$ nano q6_text_tool.sh
vboxuser@geetanjali:~/Lab2$ chmod +x q6_text_tool.sh
vboxuser@geetanjali:~/Lab2$ echo -e "hello world\nhello bash script coding" > test.txt
vboxuser@geetanjali:~/Lab2$ cat test.txt
hello world
hello bash script coding
vboxuser@geetanjali:~/Lab2$ ./q6_text_tool.sh test.txt hello
---- File Metrics ----
2 6 37 test.txt
--- Keyword Search For 'hello' ---
Keyword found! Matches shown below:
1:hello world
2:hello bash script coding
vboxuser@geetanjali:~/Lab2$

QUESTION 7.

Script to back up all .txt files from source to backup directory; debug using bash -n, bash -v, bash -x, set-x/set +x with before/after observation.

- #!/bin/bash – tells the operating system to run the script using Bash.
- echo – displays text or message on the terminal.
- read – displays the prompt and stores user's input in a variable.
- ${#files[@]}- Returns the number of elements in the array.
- for file in "${files[@]}" - Loops over each file path; quoted array expansion preserves spaces in filenames.
- cp -v -- "$file" "$dest/" - Copies each file to the backup directory; -v prints what was copied; -- ends option parsing.
- Is -I -- "$dest" - Lists backup directory contents in long format, showing details such as permissions, size, and names.
- bash -n backup.sh - Syntax check only; does not execute the script. No output usually means no syntax error was found.
- bash -v backup.sh source backup - Prints script lines as Bash reads them, then executes them.
- bash -x backup.sh source backup - Prints expanded commands as they execute; trace lines commonly start with +.
- set -x/ set +x - Turns command tracing on/off from that point within a script.
- Before / during / after tracing - before set -x, normal script output appears; between set -x and set +x, trace lines also appears; after set +x, trace lines stop.

```
vboxuser@geetanjali:~/Lab2$ nano q7_backup.sh
```




```
vboxuser@geetanjali:~/Lab2$ nano q7_backup.sh
vboxuser@geetanjali:~/Lab2$ bash -n q7_backup.sh
vboxuser@geetanjali:~/Lab2$ chmod +x q7_backup.sh
vboxuser@geetanjali:~/Lab2$ bash -x q7_backup.sh
+ src_dir=source_folder
+ bak_dir=backup_folder
+ mkdir -p source_folder backup_folder
+ echo 'Sample text'
+ echo 'Another text'
+ echo 'Starting backup process...'
Starting backup process...
+ set -x
+ for file in "$src_dir"/*.txt
+ '[' -f source_folder/file1.txt ']'
+ cp -v source_folder/file1.txt backup_folder/
'source_folder/file1.txt' -> 'backup_folder/file1.txt'
+ for file in "$src_dir"/*.txt
+ '[' -f source_folder/file2.txt ']'
+ cp -v source_folder/file2.txt backup_folder/
'source_folder/file2.txt' -> 'backup_folder/file2.txt'
+ set +x
Backup complete. Files in backup folder:
file1.txt  file2.txt
vboxuser@geetanjali:~/Lab2$
```